Medical professional

Task A: Arithmetic in medical professions

1) The table shows a patient's fluid input and output records during four hours.

time	input			output			fluid balance (ml)
	drink (ml)	liquid medication (ml)	IV fluids (ml)	urine (ml)	vomit or aspiration (ml)	drain (ml)	
09:00	200	0	125	0	0	5	320
10:00	0	10	125	300	0	25	-190
11:00	0	0	125	0	200	15	-90
12:00	150	0	125	0	0	10	265

a) Calculate the fluid balance for each hour.

Answers shown in the table.

b) Calculate the overall fluid balance for the entire 4-hour period.

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Task B: Using formulae in medical professions

1) A patient has a mass of 90 kg
They are prescribed 600 mg of a drug called rifampicin.
The drug comes vials which each have a volume of 10 ml.
Each vial contains 300 mg of the drug.
It need to be diluted up to 500 ml for the IV bag.
Once diluted, it needs to be administered continuously over 3 hours.

a) Use the formula below to calculate the dose required.

$$\text{drug dose (ml)} = \frac{\text{amount of drug prescribed (mg)}}{\text{amount of drug per vial (mg)}} \times \text{volume of each vial (ml)}$$

$$\frac{600}{300} \times 10 = 20 \text{ ml}$$

b) The 'administration rate' is the amount of the diluted drug that is given to the patient per hour from the IV bag. Calculate administration rate for this scenario.

$500 \div 3 = 166.7$ (to 1 d.p.)

c) 1 mg = 1000 mcg. How many mcg of rifampicin was prescribed to the patient?

$600 \times 1000 = 600\ 000$ mcg

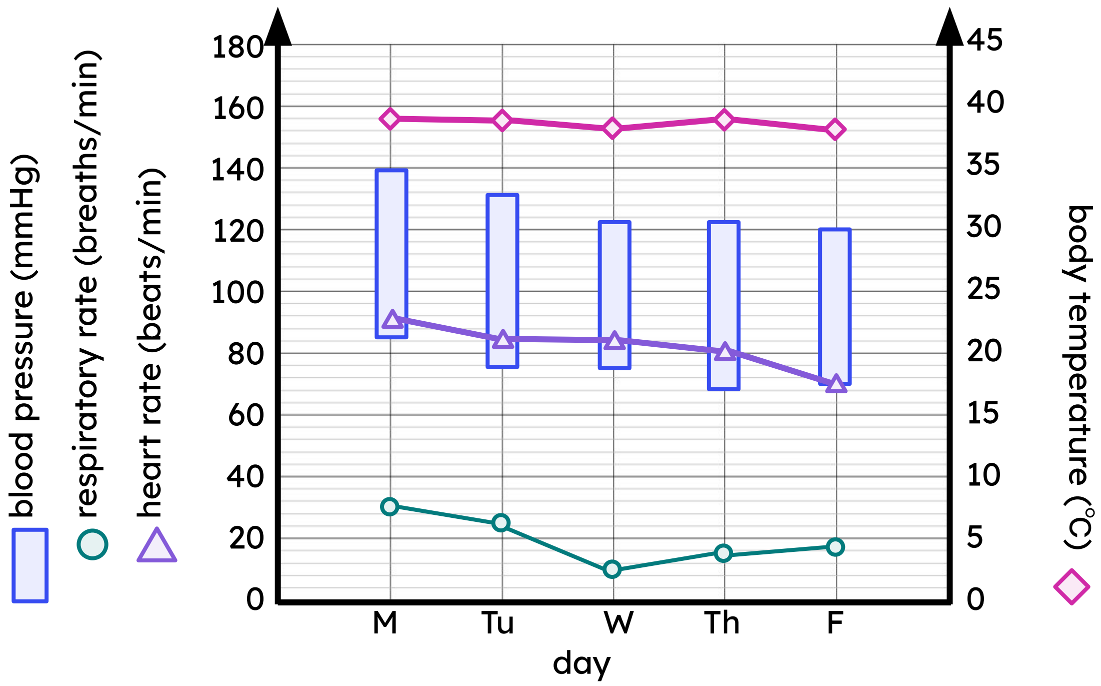
d) Use your answers from parts (b) to (c) and the formula below to calculate the mcg/kg/min settings for the IV drip.

$$\underset{\text{(mcg/kg/min)}}{\text{drip settings}} = \frac{\text{administration rate (ml/hr)} \times \text{amount prescribed (mcg)}}{\text{bag volume (ml)} \times \text{patient mass (kg)} \times 60}$$

$$\frac{166.7 \times 600\ 000}{500 \times 90 \times 60} = 37.04 \text{ mcg/kg/min}$$

Task C: Using formulae in medical professions

1) The graph shows a patient's vital signs over five days.



a) On what day was their respiratory rate at its lowest? *Wednesday*

b) What was their heart rate on Tuesday? *84 beats per minute*

c) Lucas thinks their body temperature was 155°C on Monday.
What mistake has Lucas made?

Lucas used the scale on the left rather than the scale on the right.

d) What was their body temperature on Monday? *39 °C*

e) Describe the trend in their blood pressure over the five days.

The patient's blood pressure gradually decreased over the five days.
(Or any wording that suggests that it is decreasing.)

f) A healthy blood pressure recording is around $\frac{120}{70}$

On what day(s) is the patient's blood pressure around these levels?

Friday matches exactly.

The patient's blood pressure is also close to this on Thursday and Wednesday.